8. Write a program to implement Linear Regression (LR) algorithm in python

Aim

To implement **Linear Regression** in Python for predicting a continuous numerical value.

Software Required

- Python 3.x
- NumPy
- Scikit-learn
- Matplotlib

Procedure

1. Create a dataset containing input and output values.
2. Split the dataset into training and testing sets.
3. Create a Linear Regression model.
4. Train the model.
5. Predict the test values.
6. Calculate Mean Squared Error.
7. Display the predictions.

Code

```
import numpy as np
from sklearn.linear_model import LinearRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error
X = np.array([1, 2, 3, 4, 5, 6, 7, 8, 9, 10]).reshape(-1, 1)
y = np.array([2, 4, 6, 8, 10, 12, 14, 16, 18, 20])
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = LinearRegression()


model.fit(X_train, y_train)
```

```
y_pred = model.predict(X_test)

print("Predicted Values:")

print(y_pred)

print("\nMean Squared Error:",

    mean_squared_error(y_test, y_pred))

print("\nSlope:", model.coef_[0])

print("Intercept:", model.intercept_)
```

Sample Output

```
•••  Predicted Values:
     [18.  4.]

     Mean Squared Error: 3.944304526105059e-31

     Slope: 2.0000000000000004
     Intercept: -1.7763568394002505e-15
```

Result

Thus, the **Linear Regression model** was successfully implemented and used to predict continuous values.